

LEON WANG

leonwang2026@u.northwestern.edu • (732) 925-6415
47 Ainsworth Ave, East Brunswick, NJ 08816

EDUCATION

NORTHWESTERN UNIVERSITY (GPA 3.96/4.00)

Evanston, IL

Bachelor/Master of Science in Computer Engineering

June 2026

- Relevant Coursework: Computer Architecture; ASIC & FPGA Design; Advanced Digital Logic; Microprocessor System Design; Data Structures and Algorithms; Computer System Software

RELEVANT PROJECTS

FPGA-Based Tank Duel Game

Fall 2024

- Designed and implemented a two-player “tank duel” game on an Altera DE2 FPGA board in VHDL, using VGA display, ROM memory, PS/2 keyboard controls, and an LCD screen
- Programmed continuous tank movement and collision detection and incorporated VGA Block RAM
- Developed keyboard-controlled variable speed and shooting mechanics for the tanks, allowing for dynamic real-time gameplay with score tracking on LEDs
- Simulated the game in ModelSim to verify behavior before deployment on the FPGA hardware

Pipelined RISC-V CPU

Fall 2024

- Designed a five-stage pipelined RISC-V CPU in Verilog, supporting the RV32IM instruction set
- Programmed stall and flush logic to handle long latency multiply/divide instructions and missed branch predictions, ensuring correct instruction flow
- Implemented full-forwarding to avoid RAW data hazards and increase instruction throughput
- Conducted comprehensive testing by running RISC-V assembly programs and performing detailed simulations in Icarus Verilog to validate design functionality and robustness

Path-Following Robot

Summer 2024

- Designed and built a custom path-following robot with custom PCBs, circuits, and 3D-printed parts, integrating a PIC32 microcontroller and Raspberry Pi Zero, and Raspberry Pi Camera
- Developed Python code for the Pi Zero to capture images with a Raspberry Pi camera, applying derivative edge detection and sending real-time line position data via UART to the PIC32.
- Programmed the PIC32 in C to use PWM and differential drive algorithms to control wheel motors based on line position data from the Pi Zero, achieving precise path following.

PID Motor Control Project

Spring 2024

- Programmed PIC32 microcontroller in C to implement a PID control loop for a DC motor, allowing the motor to precisely follow any user input trajectory
- Built a Python-based client to interface with the PIC32 microcontroller, allowing for user-friendly interaction, tuning, and data visualization
- Used PIC32 peripherals, such as I2C and UART, to communicate with the motor and PC

WORK EXPERIENCE

VISCOFAN

Bridgewater, NJ

Engineering Intern

June 2024 – August 2024

- Designed a PLC/Display system using a Click PLC and C-more HMI for the pressure test bench
- Programmed a Click PLC with ladder logic to control a solenoid valve based on user-defined setpoints and pressure readings to create a pressure test bench for food casing
- Built a control system for the chiller pump, which included programming a DURApulse Variable Frequency Drive with a PID loop for precise pressure control of the pump

NORTHWESTERN IMEC LAB

Evanston, IL

Undergraduate Research Assistant

December 2023 – June 2024

- Developed a Python library on Linux for millimeter-precision, real-time tracking, leveraging OpenCV to identify IR markers in video feeds and applying epipolar geometry to accurately compute the 3D positions of both cameras and markers
- Engineered the hardware system of the low-cost motion capture system for drones by modifying PS3 Eye cameras to act as IR cameras and attaching IR markers to drones

LAVNER EDUCATION

New York, NY

Information Technology Intern

June 2023 – August 2023

- Instructed high school students in programming languages, including Python, Java, and C/C++
- Responsible for troubleshooting and providing technical support to over 50 students and staff
- Set up and configured computers and software; provided technical support to students and staff

SKILLS

Languages: Verilog, System Verilog, VHDL, C, C++, Python, Java, MATLAB, R, Racket

Applications: Cadence, Genus, Quartus, ModelSim, Linux, Visual Studio, GitHub